

**University of Victoria
Engineering & Computer Science Co-op
Challenge Report**

Designing a Marine Propeller Strut

**Gregory C Marshall Naval Architect LTD
Victoria, British Columbia, Canada**

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2023-12-18**

In partial fulfillment of the academic requirements of this work term challenge

Supervisor's Approval: To be completed by Co-op Employer

This report will be handled by UVic Co-op staff and will be read by one assigned report marker who may be a co-op staff member within the Engineering and Computer Science Co-operative Education Program, or a UVic faculty member or teaching assistant. The report will be retained and available to the student or, subject to the student's right to appeal a grade, held for one year after which it will be deleted.

I approve the release of this report to the University of Victoria for evaluation purposes only.

Signature:  _____ Position: *Mechanical Specialist*

Date: 2024-02-14

Name (print): *Matt Dilay* E-Mail: *matt@gregmarshalldesign.com*

For Gregory C. Marshall Naval Architect LTD

Territorial Acknowledgement

I acknowledge with respect the Lekwungen people on whose traditional territory the University of Victoria stands, and the Songhees, Esquimalt and W̱SÁNEĆ peoples whose historical relationships with the land continue to this day.

Letter of Transmittal

Adam Geggie-Bowker
245 Meadowbrook Rd
Victoria, BC, V9E 1J5
January 6th, 2024

Monty Raisinghani
University Of Victoria
3800 Finnerty Rd, Victoria,
CA V8P 5C2

Dear Mr. Raisinghani

Please accept the following Work Term Report entitled “Designing a Marine Propeller Strut.” This report is the result of work completed at GREGORY C. MARSHALL NAVAL ARCHITECT, LTD. During my first work term as a University of Victoria Student, I engaged with the design of mechanical and electrical systems for luxury pleasure yachts.

I worked closely with a tight-knit team of engineers and naval architects to produce 3D designs and 2D construction documents in which outlined the design, outfitting, and overall specifications for shipyards to construct these vessels. Over the course of this work term, I was given the opportunity to learn about many mechanical/electrical systems as well as their designs. Some of the systems I worked with are as follows: fresh water system design, black water system design, gray water system design, ventilation design, HVAC system design, propulsion system design, along with a host of other systems. On the electrical side I was introduced to load calculations, electrical circuit design, paneling, ect.

Overall, the invaluable experience this work term has provided me will serve me not just in my future work terms, but for my further career as well. This report will focus on one of my tasks during my work term, which was design, modeling, and drafting of the marine propeller strut for the propulsion system of a hybrid motor yacht. I would also like to acknowledge and give special thanks to Matt Dilay who was my supervisor and mentor during my term, as well as all the other team members.

Sincerely,

Adam Geggie-Bowker

V00967664

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Executive Summary

This report outlines the process of designing a marine propeller strut for a 110 ft pleasure yacht. The marine propeller strut is an essential component of the propulsion system of a yacht. As a vital constituent of the yacht's propulsion mechanism, the marine propeller strut plays a pivotal role in providing essential stability for supporting the marine shaft and its accompanying propeller, housed within the hull's tunnels.



Figure 1. A P-Strut [8]

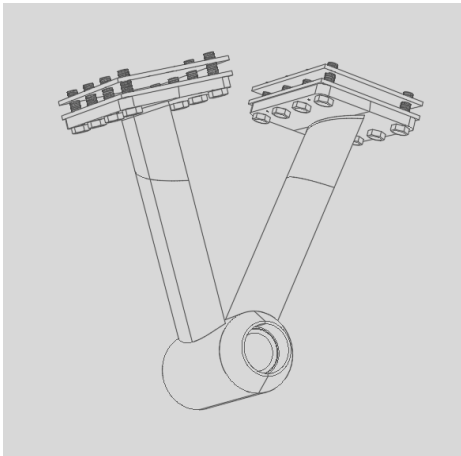


Figure 2. A V-Strut.

The placement, shape, and alignment of the marine struts can heavily influence the performance of a propulsion system. Designing struts to mitigate vibration and drag is very important and will be discussed in this report.

There are two basic designs which all marine struts are largely based on. The P-strut and the V-strut (sometimes also referred to as a A-Strut).

The P-strut (figure 1.) functions as a single strut down the centerline of the marine shaft. They are suitable for many typical marine propulsion systems, however, they are often less stable and require more material in larger vessels to meet the shafts stability and stiffness requirements. The shape allows for less overall drag in the system. The industry has moved away from these struts due to the difficulty of insallation when compared to a V-strut.

A V-strut, being the other base design, consists of two struts forming a 'V' where they then support the marine shaft. They are known for having superior stability and are ideal for contemporary vessels. These struts can be more complex to design and are more challenging to maintain as there are twice as many struts to clean, repair, or replace. However, the V-strut shows a superiority to the P-Strut by offering greater stability and stiffness using far less material. This is ideal as minimizing structure and material all while maximizing vibration and noise reduction. These aspects make the V-strut an ideal candidate for pleasure yachts.

The criteria considered for the selection of the strut design was the appearance, cost, performance, vibration mitigation, longevity, stability, ease of installation, as well as maintaining general industry

standard. During the design process analysis was performed on many different aspects of the materials as well as the positioning and attachment of the strut to the vessel.

The general recommendation for pleasure yachts is to use a V-strut design as they have become recognized as the industry standard for this size and type of vessel. Providing superior propulsion system support, and stability, this report outlines the design considerations for a custom V-strut on a 110 ft Pleasure Yacht. The goal being to design a hydrodynamic V-strut that will be secured seamlessly and affixed to the yacht's hull, ensuring both efficiency and structural integrity that effectively minimizes vibrations within the propulsion system, promoting smooth and stable and efficient yacht operation.

This report reviews the myriad of technical considerations put towards the design and development of the marine V-strut for the propulsion system of a 110 foot yacht undertaken during a CO-OP work term at Gregory C. Marshall Naval Architect LTD. The report will provide conclusions for creating a robust and efficient support system that adheres to ABS standards. The report recommends that while advancements in V-strut design enhance marine vessel propulsion, collaboration with industry leaders and adoption of emerging technologies like 3D printing offer opportunities for further optimization and innovation. Incorporating feedback from shipwrights and end-users ensures practical alignment and user satisfaction. Embracing a commitment to these innovations and cutting-edge design solidifies Gregory C Marshall Naval Architect LTD's leadership in the industry.

Glossary

Aft: rear or back portion of a vessel. It is the opposite of "forward" and is used to describe the direction or location toward the stern (rear) of the ship.

Hull: The hull of a ship is the main body or shell of the vessel. It is the watertight enclosure that floats on the water's surface and provides structural support to the entire ship. The hull is typically made of metal (such as steel or aluminum), fiberglass, or wood, depending on the type and size of the vessel.

Tunnels: Cavities in the hull housing ship propellers and its constituent components.

Cutless bearing: A type of bearing used in marine propulsion systems. It's a cylindrical sleeve made of resilient material, like rubber, fitted into the vessel's hull to support the propeller shaft. It allows smooth rotation while preventing water entry.

Propeller Shaft: The marine shaft, also known as the propeller shaft or simply the shaft, is a mechanical component that transmits power from the ship's propulsion system (such as an engine or motor) to the propeller. It extends from the engine room through the hull and connects to the propeller at the stern of the vessel.

P Strut and V-Strut: These are structural components used to support the propeller shaft and provide stability to the propulsion system. The P strut is shaped like the letter "P" and is mounted vertically on the hull, supporting the aft end of the propeller shaft. The V-strut is shaped like the letter "V" and is mounted diagonally on the hull, supporting the forward end of the propeller shaft. Both struts help to reduce vibration and ensure smooth operation of the propulsion system.

Press Fit: Press fit is a mechanical assembly technique where two cylindrical components, typically with slight interference, are joined together by applying axial force to insert one part into the other. The interference fit between the mating parts creates a secure and permanent connection, enhancing load-bearing capacity and preventing relative motion. This method is commonly used in machining, where precision and tight tolerances are crucial for achieving a snug fit. Press fits provide stability, resist rotational forces, and are widely employed in various industries, including manufacturing and engineering.



Introduction

Background

This project was undertaken during a CO-OP work term at Gregory C. Marshall Naval Architect, LTD.

Gregory C. Marshall Naval Architect LTD. is a distinguished yacht design firm with a rich history spanning over 35 years. Founded by Greg Marshall and his business partner, naval architect Gordon Galbraith, the company has earned a prominent position in the global naval architecture landscape. Greg's passion for boats since childhood, coupled with early mentorship from renowned yacht designer Bill Garden, laid the foundation for his successful career.

The firm's breakthrough came with the design of the Pacific Mariner 65, followed by acclaimed projects like the Westport 112, 130, San Juan 38, and 48. Recognized as leaders in creative yacht design and naval architecture, the team has garnered numerous awards for their innovative work, including designs like the Calixas, VVS1, and BigFish [7].

Based in Victoria, BC, Canada, the company has established itself as a trusted partner for clients worldwide, showcasing a commitment to excellence that extends into the future of yacht design.

I worked in the mechanical and electrical systems design sector of the company. The job involved creating 3D models and designs of various integral yacht systems in Rhinoceros™ and the subsequent creation of construction documents using CAD software. These construction documents outlined the different components and overall top level design of the systems.

Objective:

The marine propeller strut is an essential component of a yacht propulsion system. The strut provides the integral support and stability necessary for the cutless bearing, marine shaft and propeller to function. This report outlines the design consideration made for a custom marine propeller strut to be built onto a 110 ft motor yacht. It will delve into a comprehensive construction process of the strut and all of the pivotal design considerations made in dictating the marine strut's geometry, positioning, and attachment. The primary focus remains on highlighting the key design principles and methodologies employed in the overall construction.

Problem:

While designing a propulsion system, designing a support system that minimizes vibration and maximizes stability is imperative in ensuring efficiency and longevity.

Goals:

The overarching objective is to engineer a marine V-strut that not only complies with ABS standards but also sustains an optimal functionality throughout the yacht's designated lifespan.

The goal statement:

To design a hydrodynamic V-strut that will be secured seamlessly and affixed to the yacht's hull, ensuring both efficiency and structural integrity.

Develop the V-strut in a manner that effectively minimizes vibrations within the propulsion system, promoting smooth and stable yacht operation.

Constraints:

In the endeavor to create a marine strut that meets the highest standards, several constraints necessitate careful consideration. Foremost among these is the imperative to meet the exacting specifications outlined by the American Bureau of Shipping (ABS), ensuring strict adherence to industry regulations and safety standards.

Additionally, the design process must adhere to rigorous material and bonding standards to uphold the strut's structural integrity and durability. Conforming to industry norms and safety protocols, the size and specifications of bolting and attachment points impose specific limitations on the design, safeguarding optimal performance and reliability throughout the yacht's intended lifespan.

This report delves into the construction process and the design considerations made during the design of the marine strut. Specific analysis on stress, tearing, hydrodynamics, and other technical intricacies are not covered in this report, instead the report will primarily focus on highlighting the key design principles and methodologies. Throughout the report we will point to various cases where further analysis could have been done, but will not be considered for this report.

Discussion

This discussion will overview the different requirements and design decisions that went into the marine strut. The task was to design a marine strut that met all regulatory specifications. Minimizing vibration and maximizing stability all while meeting any standards required.

The actions I took to undertake the development of the design were to consider two key elements. The first is the design of the strut itself, and the second is the positioning and attachment of the strut to the vessel. In order to determine the development of the strut we considered the dimensions, the shape, the material required. Then in considering the position and attachment of the strut required looking at how the strut affects water flow (ie the hydrodynamics) and how it can cause vibration in the system. Also explored was the type of bearing and its attachment to the strut, as well as the fastening, and faring of the strut to the hull. The final key part of the process was the creation of the construction document for shipwrights to utilize in the successful creation and installation the strut.

Determining the Design of the Strut

As noted above there are key advantages of the strut design. In looking at the best specific design for a vessel of the strut there are key considerations, which are discussed with examples of analysis provided below.

Dimensions of the Strut:

Some considerations important in determining the dimensions of the strut include the vessel size and dimension, the compatibility with the propulsion system, with considerations to the loads and operating conditions the vessel will encounter.

For the dimensions of the vessel in question and associated strut, it was recommended to use the standards set out by ABS (American Bureau of Shipping [3]) for V-struts which are as follows:

Shaft Diameter = 3 ½"

Strut barrel Thickness > 20% Shaft Diameter =>

Strut barrel Thickness > .20 x 3.5" = 0.7"

Minor Arm Strut Thickness > 51.5% Shaft Diameter =>

Minor Arm ST > .515 x 3.5" = 1.803"

Major Arm Strut Thickness > 322% Shaft Diameter =>

Major Arm ST > 3.22 x 3.5" = 11.27"

Strut Length < 1060% Shaft Diameter =>

Strut Length < 10.6 x 3.5" = 37.1"

The provided ABS standards serve as a critical requirement for determining the dimensions of the marine strut, ensuring structural integrity and optimal performance. With a marine shaft diameter set at 3 ½ inches, the strut barrel thickness is established at a minimum of 0.7 inches, guaranteeing sufficient strength for the supporting structure. Further specifications dictate that the minor arm of the strut thickness must exceed 1.803 inches, while the major arm strut thickness should surpass 11.27 inches, both relative to the shaft diameter. These constraints prioritize robust construction to handle mechanical stresses in marine environments. Additionally, the maximum allowable strut length is capped at 37.1 inches, limiting the overall dimensions for practical and efficient integration. Adhering to these ABS standards ensures that the marine strut not only meets industry regulations but also guarantees a durable and reliable component for marine propulsion systems.

Shape:

Utilizing the dimensions found above as bounds one can then design the overall shape of the strut. Some of the considerations for shape are the hydrodynamic efficiency, the load distribution on the strut, the structural strength, and the integration of the hull design, as well as manufacturing considerations. Below we examine the considerations of shape in terms of hydrodynamic efficiency.

An rough NACA airfoil cross section for the column is more efficient for water flow than a simple oval-shaped or square shaped cross section due to the airfoil d sections' streamlined and more aerodynamic design [5]. The section minimizes drag and turbulence as water passes over the surface, reducing resistance and allowing for smoother flow. The gradual tapering of the section helps maintain a more consistent and uniform flow profile, preventing the formation of eddies and turbulence that can occur in more spherical or square sections. Additionally, the section promotes efficient energy transfer, ensuring that the water moves with less effort and more effectively through the column. Overall, the section design optimizes hydraulic performance, making it a preferred choice for applications where efficient water flow is crucial.

These are some simple 2D fluid simulations plotting high-speed water's curl passing over a 2D surface[1]. Note these are not robust simulations, they are simply demonstrating the intended design's effect. Doing robust computational fluid dynamic's (CFD) would potentially lead to improvements in the cross section design, however out of scope for this report.

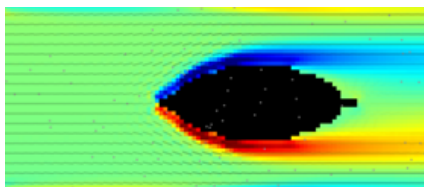


Figure 3. 2D fluid simulation A

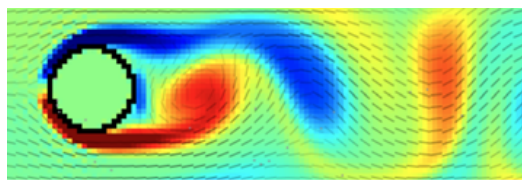


Figure 4. 2D fluid simulation B

The Final Shape and Dimension Of the Strut was then created to consider the above factors:

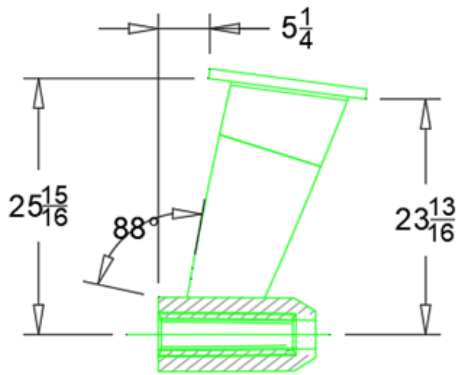


Figure 5.

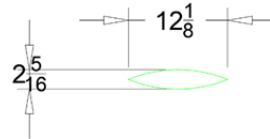


Figure 6.

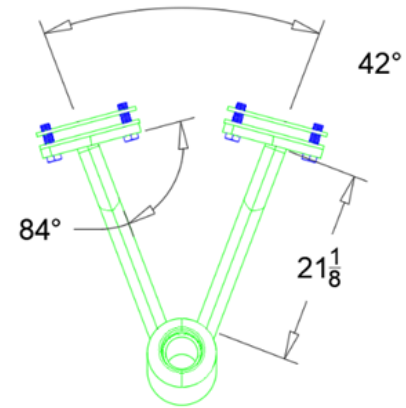


Figure 7 .

Material:

Stainless steel alloys, particularly those used in fabrication, are often deemed as an optimal material for marine propeller struts due to their exceptional corrosion resistance, high strength, and excellent machinability [6]. In the marine environment, where exposure to seawater and harsh conditions is constant, corrosion resistance is of paramount importance. Stainless steel alloys are synonymous with the ability to withstand corrosion and biofouling, ensuring the longevity and durability of the strut. Additionally, the high strength of stainless steel alloys enables them to handle mechanical stresses associated with marine propulsion systems, providing robust support for the propeller.

The excellent machinability of stainless steel alloys also allows for precise manufacturing, contributing to the production of well-engineered and efficient marine components. Overall, the combination of corrosion resistance, strength, and machinability makes fabricated stainless steel a superior material choice for marine propeller struts, offering similar benefits as copper alloys.

Attaching the Strut

In order to get optimal performance from the strut key factors are important to consider in terms of its attachment to the hull and incorporated into the design process. These factors include positioning, the load distribution of the strut, reinforcement to support load, ensuring materials are compatible, the welding and fastening methods and of course again the hydrodynamics of the propulsion system. Below we delve into considerations for positioning, attaching with a water lubricated bearing, attaching the strut pads to the hull, bolting and fibreglass faring that were part of this design process.

Positioning Of Strut Legs:

The angle of attack of the struts were positioned to be forward. Based on a typical design the struts are angled forward towards the front of the ship. This is for increased stability and allows the struts to effectively cut through the water. The angle allows for a smooth energy transfer as water passes down it and reduces the amount of force on the strut.

Collisions between propeller blades and vortexes created by water passing over the struts can induce vibrations in the propulsion system. Undesirable, this is something that needs to be minimized according to the design goals. One way to achieve this is by allowing for a larger clearance between the struts and the propellers, which would give the vortices time to disperse back into laminar flow before colliding with the propeller. However, given the position of the cutless bearing on the propeller shaft and the dimension restriction in length and angle, the total clearance achieved will still cause some turbulence by the time it reaches the propeller blade.

In the case of the Michigan 4-blade propeller being used for this boat, each propeller blade is positioned at 90 degrees relative to one another. Collisions with vortexes created by the struts are inevitable due to the propeller's location being directly behind the strut. *There is a subtle risk here, if the struts are angled at 90 degrees relative to each other, the vibration will be doubled.*

This can be reasoned by considering by as each of the vortices travel perpendicular (90 degrees) to each other they will reach the propellers at the same time. Therefore it must follow that if one struts vortex was going to strike a propeller, then the other strut's vortex would also collide with another propeller blade. This effectively **doubles** the force being applied during any given vortice strike.

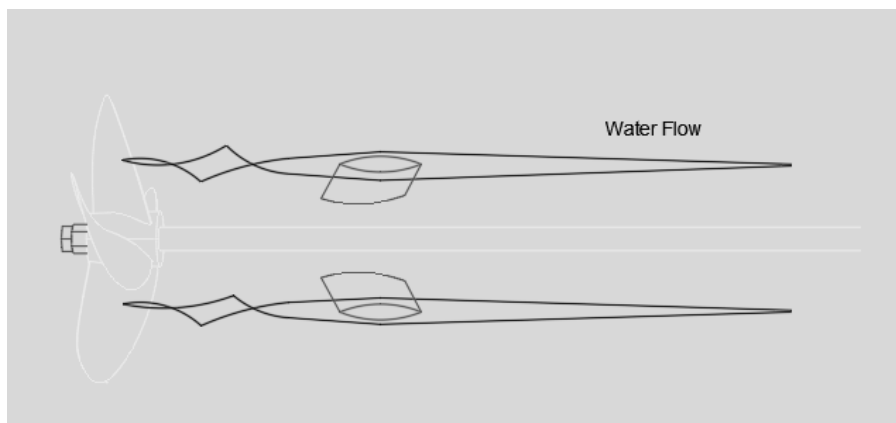


Figure 8. Water Flow

Conversely if the angle between the struts is too small, the structure lacks the necessary span to support loads applied to the propeller shaft over its lifetime. The strut centerlines total angle in respect to each other must be at a minimum of 40 degrees to each other [3]. Something else to note was the distance between the two struts when terminating at the bearing. If the strut bases are too close to each other it creates a high-pressure zone as water flows through it which creates turbulence in the water as it passes through the propeller. Ensuring adequate space between them is essential.

With the following in mind, it was decided to angle the struts at 42 degrees relative to one another, this was within our bounds for maintaining stability and far under the 90 degree angle to be avoided. The strut bases were also adequately separated to alleviate any high pressure zones that may occur.

These were the design considerations taken into account when designing the position and angle of the strut.

Attaching to a Water Lubricated Rubber Bearing:

A Water Lubricated Rubber Bearing is a type of marine bearing commonly used in propulsion systems. This self-lubricating bearing is designed to eliminate the need for oil or grease by utilizing water as a natural lubricant. The bearing consists of a rubber lining with grooves, typically made from a specialized elastomeric material, that surrounds the propeller shaft. As the shaft rotates, water flows through the bearing, providing lubrication and cooling. This design reduces friction and wear on the shaft, enhances durability, and minimizes maintenance requirements[9].

The marine struts are to be press fit to the bearing providing the strength to last beyond the yacht's intended lifetime. Grub screws will then be installed as an additional safeguard against any unforeseen failure. The attachment will be able to handle any stresses applied to it in its marine environment. The press fit should be done to typical machining standards.

Attaching the Strut Pads to the Hull:

The precise attachment of the strut pads to the hull is extremely important. Correct placement and alignment of the strut's pads insure ensure an equal distribution of the loads across the bolts and struts. The struts are to be mounted to the underside of the hull in the correct position such that they can be bolted through the skin of the hull. The strut pads will be made in three pieces.

The first piece is the strut pad's in which the strut legs are welded to. These 1 inch steel pads provide the area and the mechanical strength required for through bolting. The pad must roughly sit perpendicular to the hull tunnels curvature. Excess space under the pads are filled with CHOCKFAST® ORANGE to fill in any gaps, and correctly align the pads positions.

CHOCKFAST ORANGE (PR-610TCF) is a unique, two-component, 100% solid casting compound that has been specifically engineered for chocking or grouting applications. It's designed to endure harsh marine and industrial settings where it's exposed to high levels of physical and thermal shock. This compound doesn't shrink and boasts exceptional impact and compressive strength [2].

PR-610TCF has received approval or acceptance for marine use by numerous major regulatory agencies worldwide, including the American Bureau of Shipping, Lloyd's Register, Bureau Veritas, Det Norske Veritas, and Germanischer Lloyd.

Years of successful application have demonstrated that using PR-610TCF is not only a more effective but also a less costly approach to achieving and maintaining accurate equipment alignment in extreme conditions. Before placing in the strut pad however the hull tunnels must be recessed and sanded down for the strut pads to be properly inlaid into the hull. This recess will be faired over.

Bolting:

Ensuring that the bolts and bolt plates were the correct size to deal with all stresses applied on the strut was imperative. Per strut pad, there are 8 bolts. This totals to 16 bolts per strut. The bolt chosen for this application was the Super-Corrosion-Resistant 316 Stainless Steel Hex Head Screw. The hex head bolt is a high-strength fastener made from Grade 8 steel. It has a 1"-8 thread size and is 6" long [4]. The screw is fully threaded with a coarse thread spacing, which is the industry standard. This screw is designed for high-stress applications such as valves, pumps, motors, and automotive suspension systems. It is at least 25% stronger than medium-strength steel screws. The nut chosen to accompany is a grade 8 nut made of Super-Corrosion-Resistant 316 Stainless Steel, featuring a 1"-8 thread size and coarse thread spacing. It is about 25% stronger than medium-strength steel nuts and resists corrosion in wet environments..

BOLTS DIMENSION CALCULATION [3,6].

In adherence to the Maximum Stress Theory, failure is determined by whether the maximum stress in a material equals its failure strength under tension or compression. For stainless steel 316L, yielding failure transpires when the yield strength is attained, while fracture failure occurs upon reaching the ultimate strength. Given the uniformity of stainless steel 316L for both struts and bolts, the effectiveness of the bolt pattern design is validated when it delivers 100% of the strut strength. This ensures alignment with the principles of the Maximum Stress Theory and compliance with the material's strength criteria.

In this way:

$$\sigma_{ADM \ STRUT} = M / ZSTRUT \quad (a)$$

$$\sigma_{ADM \ BOLTS} = M / ZBOLTS \quad (b)$$

$$\sigma_{ADM \ STRUT} = \sigma_{ADM \ BOLTS} \quad (c)$$

$$ZBOLTS / ZSTRUT = 1 \quad (d)$$

$$IBOLTS / DBOLTS = ISTRUT / DSTRUT \quad (e)$$

$$ZSTRUT = 18670.94 \text{ in}^3$$

To calculate the section modules for pattern consider:

$$D = 5.11811 \text{ in (distance from bolts centerline to strut centerline)}$$

$$\text{Bolt Size} = \text{M24}$$

$$\text{Minor Thread Diameter} = 0.8563 \text{ in}$$

$$\text{Minor Thread Diameter Area (A)} = 146.3252 \text{ in}^2$$

$$\text{Bolt Pattern (N)} = 4 \text{ per side, } 5.11811 \text{ in offset from strut centerline}$$

$$\text{ZBOLT} = \text{IBOLTS} / D = [(A \times D^2) \times N \times 2] / D = 17213.54 \text{ in}^3$$

As stated in (d):

$$\text{ZBOLTS} / \text{ZSTRUT} = 18670.94 / 17952.83 = 1.04$$

=> Bolt pattern is 4% stronger then the strut.

Fiberglass Fairing:

Following the secure fastening of the struts to the hull, a critical phase in the installation process revolves around the application of fiberglass fairing. This step, while not aimed at achieving water-tightness, is primarily designed to minimize drag and enhance the vessel's hydrodynamic performance. The versatile role of fiberglass fairing encompasses not only the creation of an aesthetically pleasing appearance but also a paramount contribution to reducing drag and optimizing the overall stability of the vessel.

Beyond its cosmetic function, fiberglass fairing plays a crucial role in minimizing drag and ensuring a smooth, streamlined integration of the struts into the hull's surface. This integration minimizes turbulence, optimizing the vessel's hydrodynamic efficiency—a crucial aspect in marine applications where efficiency and stability are of utmost importance.

The fairing process involves applying layers of fiberglass material, often in the form of a resin matrix reinforced with fiberglass cloth or mat, over the strut pads and the surrounding hull area. These layers are meticulously laid and smoothed, creating a uniform surface that not only enhances the vessel's structural support but also fosters a streamlined and polished appearance.

Once the fiberglass fairing has cured, the surface undergoes meticulous sanding to achieve a smooth finish. This step is vital for attaining the desired hydrodynamic performance and ensuring the fairing seamlessly blends with the rest of the hull.

Subsequent to sanding, the fairing area is prepared for painting. The choice of paint is not only selected for visual aesthetics but also for providing additional protection against environmental factors such as UV radiation, saltwater, and abrasion.

A final inspection rigorously verifies the quality of the fiberglass fairing, checking for imperfections and ensuring the overall integrity of the installation. The meticulous application of fiberglass fairing not only enhances the vessel's performance by reducing drag but also contributes significantly to its overall visual appeal.

Create construction documents outlining the positioning.

The final step in the design process is creating construction documents. When creating the documents outlining the construction of this strut, various 2D sections and views were made. The models were then labeled and measured to scale. The intended use of the documents is for any shipwright to have access to all information required to successfully install the designed strut. The final step was documenting, and annotating which ensures that all the design steps taken are understood, and given any sort of issue with the design, is able to be diagnosed via the documentation. The details of these documents will not be covered in this report.

Conclusions

In conclusion, this project focused on the comprehensive design and construction process of a custom marine propeller strut for a 110 ft motor yacht at Gregory C. Marshall Naval Architect, LTD. The objective was to engineer a hydrodynamic V-strut that not only complies with ABS standards but also ensures optimal functionality throughout the yacht's lifespan. The report delved into the dimensions, shape, material, and attachment considerations, with a primary emphasis on key design principles.

The dimensions of the marine strut were determined in accordance with ABS standards, ensuring structural integrity and optimal performance. The shape was carefully designed for hydrodynamic efficiency, with a focus on minimizing drag and turbulence through the use of a rough NACA airfoil cross-section. Stainless steel alloys emerged as the optimal material choice for their corrosion resistance, high strength, and machinability.

The attachment of the strut to the yacht's hull involved positioning, water lubricated rubber bearing attachment, and careful consideration of factors such as load distribution, reinforcement, and hydrodynamics. The angle of attack for the struts was set at 42 degrees relative to each other, striking a balance between stability and avoiding vibration-induced risks. The use of a water-lubricated rubber bearing and strategic attachment of strut pads to the hull were crucial for achieving optimal performance and longevity.

Bolting was a critical aspect, with the selection of Super-Corrosion-Resistant 316 Stainless Steel bolts and nuts to ensure durability and corrosion resistance. The design calculations, adhering to the Maximum Stress Theory, validated the effectiveness of the chosen bolt pattern, which demonstrated a 4% strength advantage over the strut.

The application of fiberglass fairing after the secure attachment of struts to the hull contributed to the vessel's hydrodynamic efficiency by minimizing drag. This meticulous process not only enhanced the yacht's performance but also added to its visual appeal.

The final step involved the creation of construction documents, providing shipwrights with detailed information for the successful installation of the designed strut. These documents serve as a comprehensive guide, ensuring that the design steps are understood and any potential issues can be diagnosed.

In essence, this project successfully navigated the complexities of marine strut design, considering ABS standards, hydrodynamics, material selection, and attachment methods. The resulting design is poised to contribute to the efficiency, stability, and longevity of the 110 ft motor yacht's propulsion system.

Recommendations

Ongoing research at MICHIGAN WHEEL is pushing the boundaries of marine vessel propulsion by developing customized wake-adapted struts tailored to individual ship hulls. This innovation aims to minimize cavitations and drag[10]. This paired with the potential impact of modern materials and composites will transform boat strut design.

To stay ahead in the evolving landscape of yacht design, collaborations with industry leaders, researchers, and suppliers become crucial. Exploring advancements like 3D printing for specific components holds the promise of optimizing design and enhancing production efficiency.

Incorporating feedback from shipwrights and end-users into the design process is essential for iterative improvements. This approach ensures that the implemented solutions not only meet theoretical standards but also align with practical considerations and user experiences.

In summary, the dedication to embracing cutting-edge technologies and fostering collaborations will reinforce Gregory C. Marshall Naval Architect LTD's position as a leader in the yacht design and propulsion systems industry.

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